

A Learned Keyframe In-Betweeners Speeds Up Simulated Dexterous RL; Per-Stage Selection, Not Separate Training, Improves Skill Chaining

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Abstract—Modular motion models such as MotionBricks build motion from reusable parts, which makes them attractive for dexterous hands, whose tasks are long and whose demonstrations are scarce. Unfortunately, prior work on synergies, latent action spaces and skill chaining changes several things at once, so it is unclear where a modular gain comes from. We test each of MotionBricks’ two layers on simulated hands, with the main comparisons registered before they ran. At the motion level, a keyframe action space, target poses joined by a learned in-betweeners, leads a continuous latent that decides at the same rate at the registered checkpoint (0.59 against 0.22 success on held-out skill sequences at 1.5 M frames; difference +0.38, 95% CI [0.23, 0.52]) and reaches its training plateau sooner. Other modular action spaces, straight-line keyframes and a bank of fixed primitives, do not. At 5 M frames the data do not separate it from the latent. At the skill level, selecting the best policy for each stage completes 97 of 400 test episodes against 61 for the best single policy, in one run of the selection pipeline; training the skills separately does not help (45 of 400). The gains come from the search, faster learning with a learned in-betweeners and per-stage selection, not from separately trained modules.

Index Terms: Dexterous Manipulation, Reinforcement Learning, Learning from Demonstration, Multifingered Hands, Deep Learning Methods.

I. INTRODUCTION

Dexterous manipulation tasks are long and their demonstrations are scarce, so a learner that reuses parts is attractive. MotionBricks [1] generates character motion from reusable parts at two layers: target poses joined by a learned in-betweeners, and smart primitives that act. Modularity of this kind promises a hand that learns faster, because each part is simpler than the whole task, and performs better, because parts recombine into behaviour never seen whole.

Whether modularity keeps these promises in control is unclear. Synergy and latent action spaces report faster learning [2]–[6], and skill chaining works on dexterous hands with engineered hand-offs [7], [8]. Each of these changes the prior, the decision rate, the data or the training procedure together, so it is unclear which part of the design produces the gain.

This paper separates the parts. A flat policy that knows which stage it is in can, with enough capacity, represent what a modular policy represents, so a modular gain is expected in how a solution is found rather than in which solutions exist. We therefore test two routes by which modularity changes the search: an action space built from poses and a learned in-betweeners, compared with a continuous latent at the same decision rate; and modular deployment, which lets each stage’s policy be selected on its own, compared with separate training on the same data.

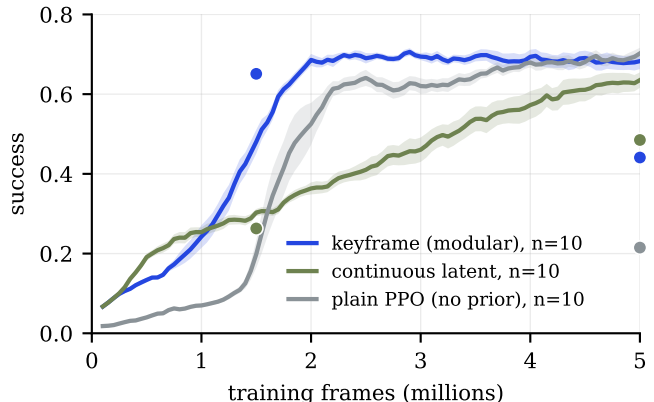


Fig. 1. Keyframes with a learned in-betweeners learn faster. On the training sequences the keyframe interface (blue) reaches its plateau by 2 M frames, while plain PPO (grey) and the continuous latent (green) take longer; by 5 M frames the three are within 0.06 of each other. On held-out sequences (dots) it leads the latent at 1.5 M frames; at 5 M the two are not distinguishable. Lines: training success, mean and standard error over 10 RL seeds; dots: the same seeds (Section IV reports all 20 seeds at 1.5 M).

Both routes pay. At the motion level, a keyframe interface in MotionBricks’ sense leads a continuous latent that decides at the same rate on held-out skill sequences at 1.5 M frames (+0.38) and reaches its training plateau sooner; by 5 M frames the data do not separate the two (Fig. 1). At the skill level, selecting the best policy for each stage completes more episodes than the best single policy; training the skills separately does not (Fig. 3).

Our contributions are:

- 1) **Faster learning from a learned in-betweeners.** At the same decision rate, keyframes joined by a learned in-betweeners lead a continuous latent at the registered checkpoint and reach the training plateau sooner than the latent and plain PPO (post hoc); straight-line keyframes and fixed primitives do not (Section IV).
- 2) **Per-stage selection, not separate training.** A design that separates the two shows that selecting each stage’s policy on its own raises success in our run, and that separate training on the same demonstrations does not (Section V).

II. RELATED WORK

Priors as action spaces. A learned motion prior turns a high-dimensional hand into a smaller search space. LAMP [9] uses a continuous latent prior as the action space for residual RL, and DexSynRefine [10] grounds motion

primitives with residual RL. Synergy, latent and dynamical-system action spaces mostly speed up early learning [2]–[6], and gains in final success appear where the joint-space baseline has not converged [11], [12]. Learned skill priors [13] and action chunking [14] shorten the decision horizon in the same way as a keyframe interface. What this line of work leaves open is how much of the speed-up is the structure of the prior and how much is the decision rate; the comparison here holds the decision rate fixed.

Hierarchy and skill chaining. Options [15] and their end-to-end learning [16], [17] formalise policies that act through sub-policies. Modular motion appears as authored primitives over a latent backbone [1], switching dynamical systems [18], [19] and mode-adaptive gating [20]. Skill chaining trains each skill to start where the previous one ends [21], terminal-state regularisation helps the next skill take over [8], and on dexterous hands chaining succeeds when the hand-offs are engineered [7]. These methods add machinery at the hand-offs, so they do not isolate what training modules separately contributes. Analyses of hierarchical RL trace most of its benefit to exploration [22] and its efficiency to sub-problems that repeat [23], which points to the search.

Policy selection. Imitation-learned policies vary widely across seeds and checkpoints, and the choice among them matters as much as how they are trained [24], [25]. That work selects whole policies; chaining pipelines select each skill, but do not measure what this selection contributes against separate training. On the relocate task used here, DAPG [26] fine-tunes a cloned policy with RL; our skill level uses behaviour cloning alone, to isolate selection from training.

III. HAND-MOTION PRIORS

All priors are trained on OakInk [27] hand motion, represented as 27 anatomical degrees of freedom (four per finger, five for the thumb, six for the wrist) projected from MANO [28]. The *continuous latent* is a variational autoencoder ($\beta = 1$ with free bits [29]) that encodes a 32-frame window into a 12-D latent per frame, with GRU encoder and decoder (2.28 M parameters) and a bounded output, so joint limits hold by construction. The *keyframe in-betweenner*, the MotionBricks-style prior, is a GRU trained on the same motion to fill the frames between two target poses. The *clip bank* is a bank of $K = 16$ fixed motion primitives mixed softly with a style code.

IV. MOTION LEVEL: A LEARNED IN-BETWEENNER LEARNS FASTER

Interfaces. Each prior becomes the action space of PPO [30], and each prior interface takes one decision every 16 frames. (a) Clip bank: a soft mixture of primitives and a style. (b) Continuous latent: the 12-D latent, held for 16 frames. (c) Keyframe, in MotionBricks’ sense [1]: a target pose, reached by the learned in-betweenner. (d) The same target pose reached by straight-line interpolation, with no learned prior. Each adds a per-frame residual of at most 3° ,

capped at $\pm 15^\circ$ in total. (e) Plain PPO sets joint targets every frame, with no prior. Every policy observes a one-hot code of the current skill and the fraction of the skill elapsed.

Task. On the Shadow Hand E3M5 [31] in MuJoCo [32], a 5 cm, 0.20 kg cube rests on the palm while a lateral acceleration of 1.5 g acts on it, as if the palm were tilted. An episode is a sequence of skills of forty frames each: hold with the acceleration along x , hold with it along y , release. Policies train on the four two-skill sequences and are scored on the two three-skill sequences they never see, over 100 episodes with the deterministic policy; an episode succeeds when the cube is held at the end of every hold and has left the hand at the end of the release.

Training. PPO runs 16 parallel environments with rollouts of 256 steps, batches of 1024, learning rate 3×10^{-4} , entropy coefficient 0.01, 10 epochs per update, $\gamma = 0.99$, $\lambda = 0.95$ and clip 0.2. Actor and critic have two hidden layers of 256 units; exploration is state-dependent, resampled every 8 steps, from an initial log standard deviation of -1 . The reward is one unit for every frame the cube is held during a hold, and a graded per-frame reward during release. Every interface uses the same settings; the interfaces share the decision rate but differ in action dimension and in the scale of their exploration noise.

Faster. At 1.5 M frames the keyframe interface succeeds on 0.59 of held-out sequences against the latent’s 0.22 (20 seeds; difference $+0.38$, 95% CI $[0.23, 0.52]$, t -interval over seeds; 17 of 20 seeds, sign test $p = 0.003$, as declared), and the lead holds on longer unseen sequences of four and five skills (0.51 and 0.48 against 0.22 and 0.15). Fig. 1 shows the same lead on the training sequences: the keyframe interface reaches its plateau of 0.69 by 2 M frames, plain PPO comes within 0.05 of it at 2.5 M, and the latent is still below it at 5 M (0.64). Over the first 2.5 M frames its mean training success is 0.40, against 0.25 for plain PPO and 0.27 for the latent (10 seeds each, Mann–Whitney $p = 10^{-4}$ for both, post hoc). Because the latent also takes one decision every 16 frames, the difference is not the decision rate. The same target poses reached by straight lines never learn to release (0 of 10 seeds), so on this task the learned in-betweenner, not the keyframes alone, carries the gain. Not every modular action space helps: the bank of fixed primitives is the weakest prior (0.08), so the gain belongs to the learned in-betweenner rather than to modularity as such. Fig. 2 shows one held-out episode.

The end point. At 5 M frames (seeds 0–9) the data do not separate the keyframe interface from the latent on held-out sequences (0.44 against 0.49; paired difference -0.04 , 95% CI $[-0.22, 0.13]$, inconclusive). It still beats plain PPO there (0.44 against 0.22; difference $+0.23$, 95% CI $[0.09, 0.36]$; 9 of 10 seeds, as declared), but plain PPO also decides every frame, so this comparison changes the decision rate as well as the prior. On the training sequences the three end within 0.06 of each other. Longer training does not help held-out success: on the same seeds the keyframe interface falls from 0.65 at 1.5 M to 0.44, as both learned priors settle on holding perfectly (0.96 to 1.00 per hold on the training sequences)

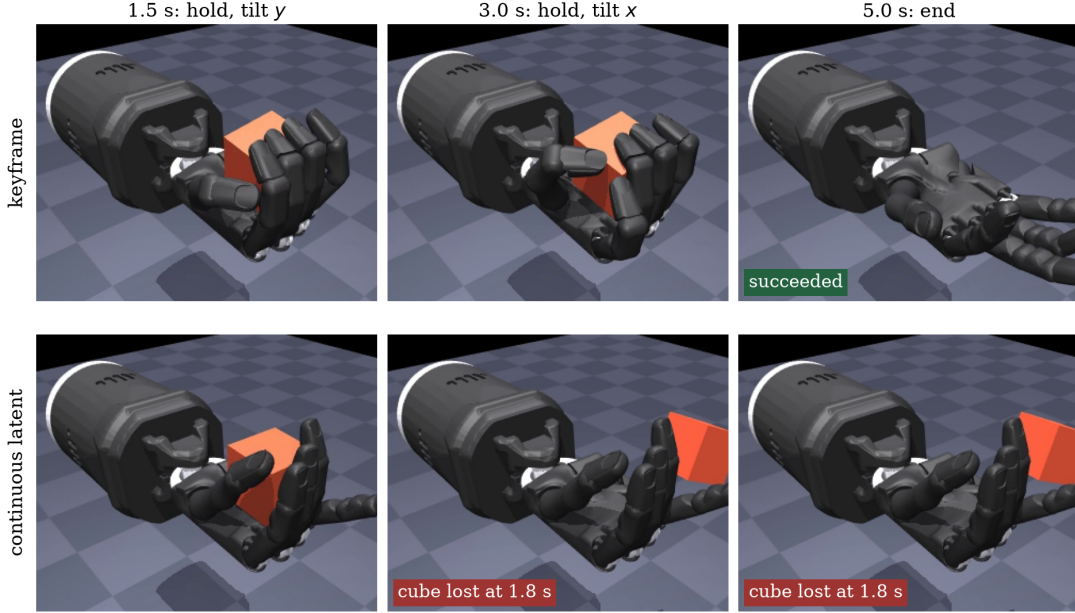


Fig. 2. The keyframe interface holds the cube through an unseen sequence; the continuous latent drops it. One held-out sequence at 1.5 M frames (hold under a y tilt, hold under an x tilt, release); top, the keyframe interface; bottom, the continuous latent, which loses the cube during the first hold. RL seed 13, whose success is closest to the 20-seed means; the first evaluation episode (of 100) in which the keyframe interface succeeds and the latent fails.

TABLE I

EACH PRIOR AS A POLICY’S ACTION SPACE: SUCCESS ON SKILL SEQUENCES NEVER SEEN IN TRAINING. THE KEYFRAME INTERFACE LEADS AT 1.5 M FRAMES; AT 5 M IT IS NOT DISTINGUISHABLE FROM THE LATENT. SEEDS: 20 AT 1.5 M, 10 FOR THE STRAIGHT-LINE AND BANK ROWS AND AT 5 M.

interface	1.5 M frames	5 M frames
plain PPO (no prior)	–	0.22
continuous latent	0.22	0.49
keyframe, in-betweeners	0.59	0.44
keyframe, straight-line	0.00	–
$K = 16$ clip bank	0.08	–

and releasing less than half the time. On three other objects (a heavier cube, a sphere and a cylinder) the early lead is smaller and not significant (pooled test, $p = 0.067$).

V. SKILL LEVEL: SELECTION, NOT SEPARATE TRAINING

Task. MotionBricks’ other layer composes smart primitives, units that act rather than poses. The test uses Adroit’s relocate [26], extended into a sparse multi-stage task: grasp the ball from the table, then carry it to a random target in the air, once or twice. The data are the 25 human demonstrations of D4RL [33], each containing one grasp and one carry, and every policy is behaviour-cloned from them; PPO from scratch never completes a carry (5 M steps, two seeds). A modular policy trains one skill per stage and chains them with a fixed sequencer; a *flat* policy is one policy for the whole task, with the stage index in its input. A stage counts as done only after its goal holds for 25 consecutive steps: with a one-step criterion, carry policies learn to fling the ball

through the target, and the next skill cannot recover it [8].

Design. The design separates selection from training, on fresh seeds and 400 test episodes never used for any choice, with 20 trainings per side. Variant A is the best of 20 flat policies, used for every stage. Variant B takes the same 20 flat policies and uses the best grasper for the grasp and the best carrier for every carry. Variant C trains 10 grasp skills (on the grasp segments of the demonstrations) and 10 carry skills (on the carry segments) and chains the best of each. A against B changes only the selection; B against C changes only the training. Every choice uses 200 validation episodes per stage: graspers start from the benchmark’s reset, and carriers start from a fixed bank of states in which a separate grasp policy holds the ball. Behaviour cloning regresses the policy’s mean action onto the demonstrated actions (mean squared error, Adam, learning rate 10^{-3} , batch 256, 20 epochs), with the same network as above and observation normalisation fitted to the demonstrations. An earlier registered comparison with five seeds per variant is at the floor at its declared length of four stages ($p = 0.45$); at two stages the modular variant wins (38 against 8 of 200, $p < 10^{-5}$), which we traced after the fact to per-stage selection. This design tests that reading.

Per-stage selection helps. In this run of the selection pipeline, selecting the best policy for each stage beats the best single policy: 97 against 61 of 400 episodes at two stages ($p = 0.002$) and 47 against 12 at three ($p < 10^{-5}$). It wins through the carry: B’s carrier completes 0.34 of carries after a grasp, against A’s 0.20.

Separate training does not. The registered prediction, that separately trained skills beat the best flat policy, fails (45 against 61, $p = 0.12$), and separate training loses to

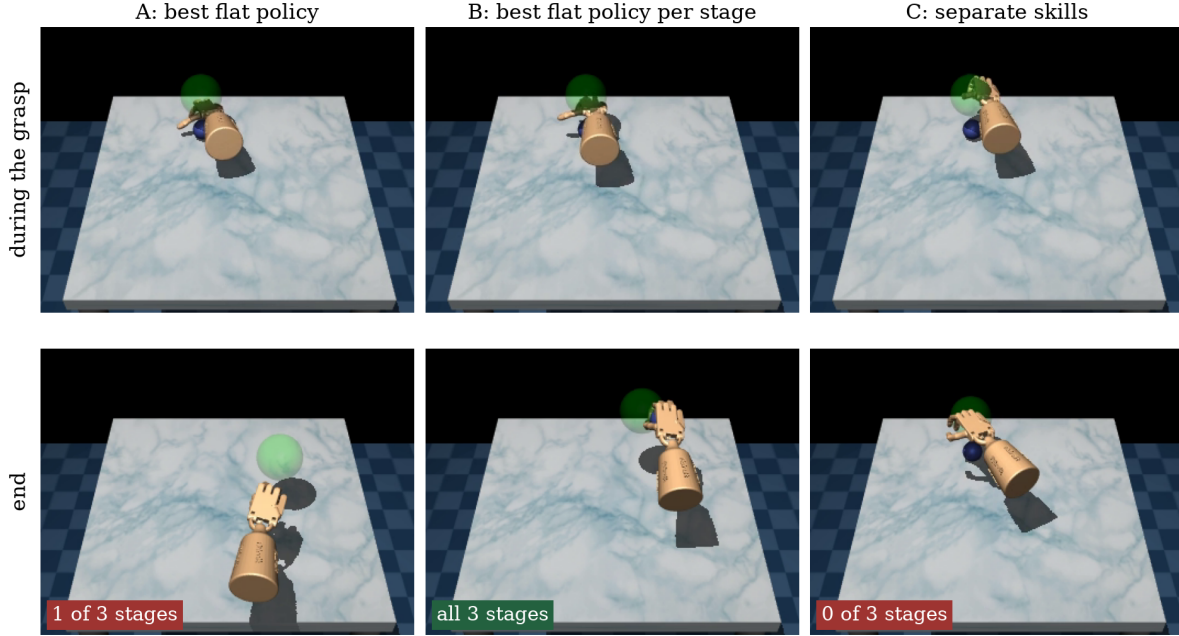


Fig. 3. Selecting per stage completes the task where the other variants stop. Same test start for the three variants (grasp, carry, carry again); top, during the grasp; bottom, at the end. B, the best flat policy for each stage, completes all three stages; A grasps the ball but never delivers it, and C never completes the grasp. One test episode, the first in which B completes every stage; B does so in 47 of 400 test episodes (Table II).

TABLE II

SELECTION, NOT TRAINING: 400 FRESH TEST EPISODES, PRE-REGISTERED. PER-STAGE SELECTION (B) COMPLETES THE MOST EPISODES, THROUGH THE CARRY. PER-STAGE RATES ARE FOR THE TWO-STAGE TASK (CARRY | GRASP: CARRIES COMPLETED AFTER A COMPLETED GRASP); THE LAST TWO COLUMNS COUNT EPISODES THAT COMPLETED EVERY STAGE. FISHER’S EXACT TEST AGAINST VARIANT A, OVER TEST EPISODES FOR THE SELECTED COMBINATION; THE PIPELINE RUNS ONCE.

variant	grasp	carry grasp	2 stages	3 stages
A: best flat policy	0.77	0.20	61	12
B: best flat per stage	0.72	0.34	97 ($p = 0.002$)	47 ($p < 10^{-5}$)
C: separate skills	0.57	0.20	45 ($p = 0.12$)	6 ($p = 0.23$)

per-stage selection (45 against 97, $p < 10^{-5}$). C loses at the grasp (0.57 against 0.77). Without selection, the declared comparison over seeds finds separately trained carry skills worse than the carry stage of flat policies (0.14 against 0.27 on validation, $p < 10^{-5}$) and grasp skills not distinguishable ($p = 0.19$); on the test episodes both are lower (grasp 0.14 against 0.23, carry 0.17 against 0.26, post hoc).

Not a matter of pool size. B chooses from 20 policies per stage and C from 10, and each variant is one selected combination. A post hoc follow-up, with its analysis written down before it ran, scores every selection rule by the product of its per-stage test rates. The product assumes a carrier’s success does not depend on which grasper precedes it; for the three selected combinations it is within 0.05 of the chained test result (A 0.20 against 0.15, B 0.26 against 0.24, C 0.10

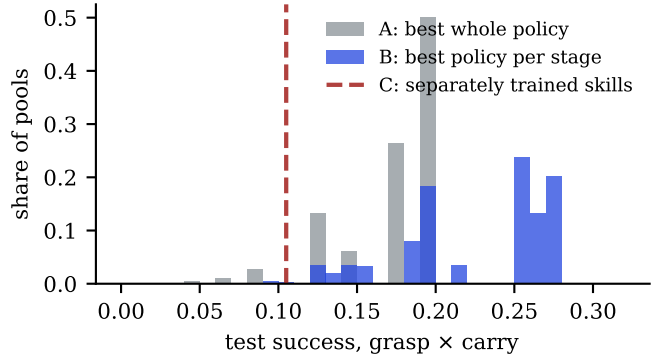


Fig. 4. Selection, not pool size. With pools matched at 10 policies per stage, per-stage selection (B, blue) lies above separate training (C, dashed line) in 99% of pools and mostly above whole-policy selection (A, grey). Each bar counts pools of 10 flat policies (184 756 in all) by the product of per-stage test rates (grasp × carry), with every choice made on validation.

against 0.11). Restricted to each of the 184 756 possible pools of 10 flat policies, per-stage selection still beats separate training in 99% of pools (mean 0.23 against 0.10). Over 10 000 redrawn pools of 10 per side, it beats separate training in 96% of draws and whole-policy selection in 82% (Fig. 4). These pools are subsets of the same 20 policies scored on the same test episodes, so they show that the result does not hinge on pool size, not that it replicates; repeating the pipeline on independent seeds would test that.

VI. DISCUSSION: WHERE THE GAINS COME FROM

Both results point to the same place. Keyframes with a learned in-betweeners pay by exploring faster; modular

TABLE III

THE FOUR CONDITIONS FOR SEPARATELY TRAINED MODULES TO PAY,
AGAINST THE TWO TASKS. NEITHER TASK MEETS (IV).

condition	motion level	skill level
(i) solution inside	partly: the grip needs 83°, OakInk and TACO [34] reach 73–76°; the residual covers the gap	yes: skills learned on the task
(ii) clean hand-offs	yes: poses only	only when enforced: a one-step stage lets carry fling the ball
(iii) something to amortise	partly: faster exploration, one task	yes: PPO from scratch never carries
(iv) flat cannot match	no: ends level with the latent	no: flat policies selected per stage do best

deployment pays by letting each stage’s policy be chosen on its own. Neither requires training modules separately, and at the skill level separate training costs skill. This is consistent with the finding that most of the benefit of hierarchy comes from exploration [22], and with synergy and latent action spaces, which report faster early learning [2]–[6]. The speed-up depends on the prior: a learned in-betweeners explores well, while straight-line keyframes and a bank of fixed primitives do not.

Per-stage selection works because each stage is validated on its own. With N trained policies, per-stage selection picks the best of each stage, where whole-policy selection picks only the best product. If a carrier’s success does not depend on which grasper precedes it, the product of the per-stage best is never below the best whole policy; the chained results above stay close to this product. Selection is therefore expected to help; what the experiment measures is how much, and that separate training does not add to it. Chaining pipelines already select each skill [7], [8]; the result here measures what that selection contributes on its own, and shows that it, not separate training, carries the gain.

When would separately trained modules pay as well? A chain of modules succeeds with probability $\prod_i P(\text{module } i \mid \text{end states of module } i-1)$, so the cost sits at the hand-offs. Four conditions, formulated during this study and offered as hypotheses, state when separate modules would pay: (i) the solution lies inside what the modules express; (ii) hand-offs are clean; (iii) there is something to amortise, such as a horizon a flat learner cannot cross or combinations it has not seen; and (iv) a flat learner given the same data does not get the same benefit. Table III maps the two tasks onto them; neither meets (iv). MotionBricks, in animation, plausibly meets all four: its goal is human-like motion and its data are human motion, it is kinematic so any pose starts the next brick, one library serves many applications, and it has 10^5 clips.

VII. LIMITATIONS AND FUTURE WORK

Everything here is in simulation, with one task per layer, and the early lead is significant on the cube only. All interfaces share one set of PPO settings, and plain PPO

also differs in decision rate; per-interface tuning and plain PPO with repeated actions would sharpen the motion-level comparison, as would held-out success along the whole learning curve. The skill-level selection pipeline runs once, so its tests count test episodes for one selected combination; repeating the pipeline on independent pools would measure its variance directly. The skill level uses behaviour cloning from 25 demonstrations and a fixed sequencer; RL fine-tuning and hand-off methods such as T-STAR [8], as well as learned skill priors and action chunking at the motion level, are natural next baselines. Longer training with interface-specific tuning would show whether the end points stay level. The four conditions are hypotheses; a task that meets all four should show a gain from separately trained modules.

VIII. CONCLUSION

On these simulated dexterous hands, a MotionBricks-style keyframe interface with a learned in-betweeners learns faster than a continuous latent at the same decision rate, and the two end at levels the data do not separate. At the skill level, selecting the best policy for each stage raises success, while training the skills separately does not. The gains come from the search, not from separately trained modules.

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